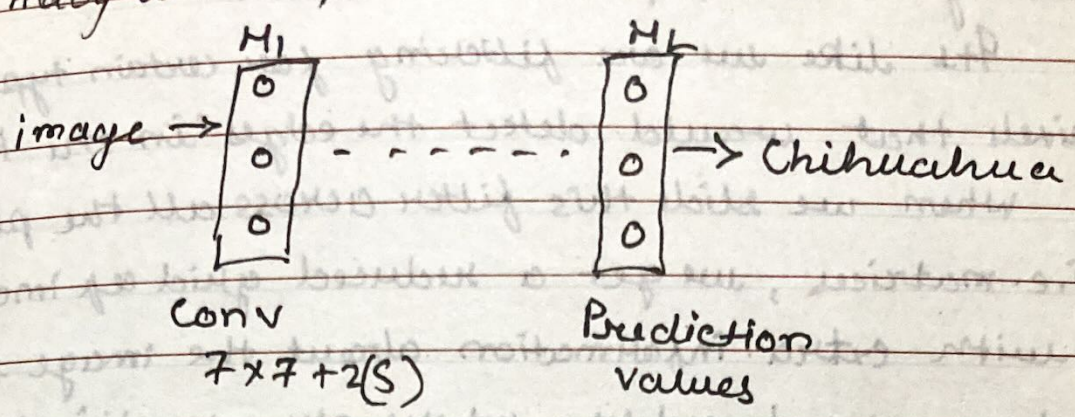


How CNNs work?

lets take an example
A dog breed predictor

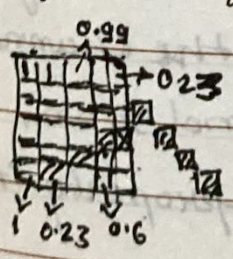


How is this happening

H_1 layer is activated by pixels in the image.
 H_2 layer i.e. last hidden layer is having the values indicating the category of dog.

Whats happening in middle exactly?

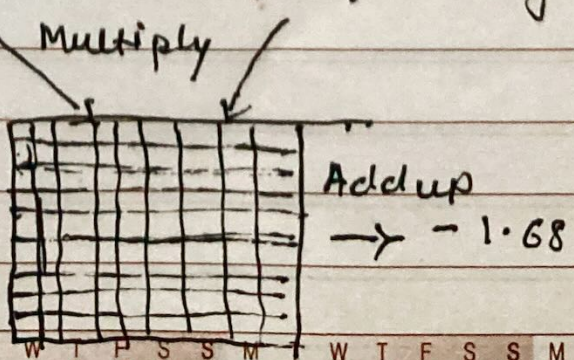
Important layer is the convolutional layer.
 In the first layer of input cells in the form of a grid of



A matrix of pixel activations



A matrix of weights for filtering



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suppose a positive value indicates pixels on top is brighter than pixels at bottom.

Its like we are filtering for certain type of pixels that would detect the edges in the image.

When we slide this filter across all the pixel grid i.e. matrices, we get a reduced grid of image with extra information about the image like where are borders, where are curves in the original image.

We also add a bias term after applying the filter, if we want a high or low value.

If result is -ve we round it up to 0.

Now we have multiple filters i.e. weight grids producing different activated grids. These activated grids are called channels.

These channels together form a new layer.

We again apply more filters to the channels to pass to new layer as input and so on.

Each of these layers detect a property in the image like ears or tail shape of dog